EART26201 | Principles of Geochemistry | Practical 1

Trace element modelling FEEDBACK

Task

This practical contains two questions about using trace element partitioning to understand igneous processes. Each question should take about 60 minutes.

I strongly encourage you to discuss your work with each other as you go.

Demonstrators will be available throughout the practical class, and I will provide a short round-up at the end of the session. Feedback will be posted on Blackboard in due course.

Please share any comments you have as we go along, and I will do my best to adjust.

I suggest typing your answers into this document and/or its associated Excel file.

Goals

At the end of this practical you should be able to:

- Calculate bulk partition coefficients
- Describe how the concentrations of compatible and incompatible elements change during melting and crystallisation
- Calculate melt fractions and trace element concentrations using the Rayleigh fractionation equation
- Create and manipulate simple geochemical plots in Excel
- Interpret multi-element diagrams to determine mantle sources of igneous rocks

Other themes relevant to this week's content are reviewed in a Blackboard quiz.

You will revisit many of these themes in EART22200 (Igneous Minerals and Processes) and EART24200 (Metamorphic Minerals and Processes) later in the year.

Before you start

Open the file *EART26201_Practical1.xlsx* in Excel.

Question 1: Using trace elements to model crystallisation processes (60 minutes)

- La Palma is an ocean island volcano in the Canary Islands in the Atlantic Ocean (28.67°N, 17.84°W). Here we will use the trace element content of volcanic rocks from La Palma to investigate magma crystallisation processes taking place beneath the volcano. Specifically, we will consider two types of alkali-rich lavas called basanites and phonotephrites. The compositions of these lavas are provided in Table 1.

Table 1: Trace element analyses of lavas from La Palma, Canary Islands. B, basanite; PT, phonotephrite (sample names from Klügel et al., 2000).

µg/g	B1	B2	B3	PT1	PT2	PT3	PT4
Cr	767	377	328	127	115	60	39
Rb	38	33	30	58	70	103	69

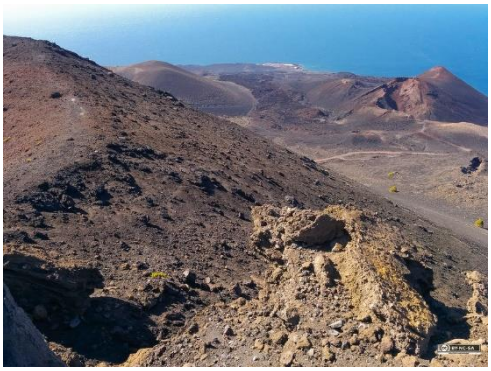


Figure 1: Volcanic landscapes associated with Teneguía cone on La Palma. (credit: weberpal)

- At equilibrium, the concentration of trace element i in a solid crystal phase (C_i^{solid}) is related to the concentration of i in the liquid from which the crystal grew (C_i^{liquid}) by the partition coefficient (D_i) as follows:

$$D_i = \frac{C_i^{solid}}{C_i^{liquid}}$$

- The four main crystal phases in basanites from La Palma are olivine (ol), clinopyroxene (cpx), hornblende (hbl) and magnetite (mgt). Partition coefficients for two trace elements, Cr and Rb, in these crystals are given in Table 2 (overleaf). If we know the proportions of these crystal phases, we can calculate the bulk partition coefficient ($\overline{D}_i$) for the whole crystallising assemblage (Lecture 1 | Part 3). Assuming that these crystal phases grew from the basanite melt in the proportions 10% ol, 40% cpx, 40% hbl, 10% mgt. The bulk distribution coefficient ($\overline{D}_i$) for trace element i in the La Palma samples is therefore given by the following:

$$\overline{D}_i = 0.1D_i^{\mathrm{ol}} + 0.4D_i^{\mathrm{cpx}} + 0.4D_i^{\mathrm{hbl}} + 0.1D_i^{\mathrm{mgt}}$$

Table 2: Solid-liquid partition coefficients for the four main crystal phases in La Palma lavas: ol, olivine; cpx, clinopyroxene; hbl, hornblende; and mgt, magnetite.

	D^{ol}	D^{cpx}	D^{hbl}	D^{mgt}
D_{Cr}	0.9	8	0.34	15
D_{Rb}	0	0.002	0.2	0

a) Using the D_{Cr} values given in Table 2 and the expression given in c), calculate the bulk partition coefficient, $\overline{D}_{Cr}$, for basanites from La Palma. You can do this in the sheet titled *Question 1* in the Excel file you have downloaded. Remember that you can insert $ before coordinates used to select cells (e.g. A1) to ensure that values do not change when copying or dragging cell contents; you can do this for columns, rows or both.

4.9

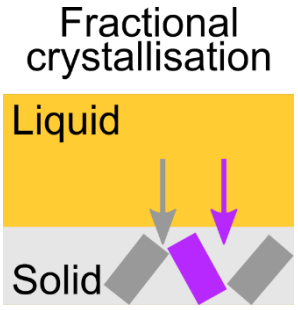
b) Using your answer to a), state whether the concentration of Cr in the crystallising assemblage is higher or lower than the concentration of Cr in the melt?

The concentration is higher in the crystallising assemblage than in the melt because Cr is compatible.

c) Assuming that the proportions of ol, cpx, hbl and mgt stay constant during crystallisation, will the concentration of Cr in the residual liquid increase or decrease?

The concentration of Cr will decrease

d) If we know the fraction of liquid remaining (F), the concentration of Cr in the residual liquid can be calculated using the Rayleigh fractionation equation, which assumes that crystals are extracted from the melt as soon as they are formed:



$$\frac{C_l^{liquid}}{C_0^{liquid}} = F^{(\overline{D}_i - 1)}$$

where C_0^{liquid} is the concentration of element i in the starting liquid, C_l^{liquid} is the concentration of an element in the residual liquid, F is the fraction of liquid remaining and $\overline{D}_i$ is the bulk partition coefficient of element i. Rearranging this equation for F gives:

$$F = \left(\frac{C_l^{liquid}}{C_0^{liquid}}\right)^{\left(\frac{1}{\overline{D}_i - 1}\right)}$$

Assume that phonotephrite melts on La Palma are generated from the most primitive basanite (B1) following the fractional crystallisation of ol + cpx + hbl + mgt in the proportions given in c). Using $[\mathrm{Cr}]_0^{liquid} = 767\ \mu g/g$ (the concentration in B1; Table 1) and $[\mathrm{Cr}]_l^{liquid} = 39\ \mu g/g$ (the concentration in PT4, the lowest Cr sample, Table 1), and the value of $\overline{D}_{\mathrm{Cr}}$ from d), calculate the fraction of the starting liquid that is represented by the phonotephrite. In other words, what is F?

F = 0.47

These calculations tell us that $[(1-F)\times 100]\%$ of the primitive basanite has been extracted as a mixture of ol + cpx + hbl + mgt crystals to leave the phonotephrite melt, which is a considerable amount of fractionation in this case. (Note that we have assumed that the crystallising assemblage does not change, which is probably an over-simplification. Also, given the large value of $D_{\mathrm{Cr}}^{\mathrm{mgt}}$, even a small error in the proportion of magnetite could propagate into a large error in the value of F.)

e) Now determine whether the calculated amount of fractionation from basanite to phonotephrite is consistent with the Rb data. First, calculate $\overline{D}_{\mathrm{Rb}}$ in the same way as you calculated $\overline{D}_{\mathrm{Cr}}$ in d). Then using $[\mathrm{Rb}]_0^{liquid} = 38\ \mu g/g$ (the concentration in B1; Table 1) and the value of $\overline{D}_{\mathrm{Rb}}$ that you just worked out, calculate $[\mathrm{Rb}]_l^{liquid}$ for the value of F that you obtained in g). Is your calculated concentration of Rb in the phonotephrite consistent with the observed concentrations in Table 1?

$\overline{D}_{\mathrm{Rb}}$ = 0.0808. The estimated value of [Rb]l is 76.3 µg/g, which is very close to the observed concentration in sample PT4 (69 µg/g).

f) Plot the concentrations of Cr (x-axis) against Rb (y-axis) as a scatter plot in Excel by first selecting the data (click on cell B4 and drag over to C10; by default Excel will make the left-hand column the x-axis and the right-hand column the y-axis) and then clicking on *Insert* → *Charts* → *Scatter* (the top-left option without any lines). Remember to include axis labels (to add axis labels click on the chart and then select *Add Chart Element* → *Axis Titles* under the *Chart Tools* → *Design* menu) and to scale the plot appropriately (you can do this by dragging the corners of the selected plot or by typing the *Size* under the *Chart Tools* → *Design* menu after selecting the chart). Notice that the data form a curve. This is characteristic of fractional crystallisation trends and makes plots of

See Excel spreadsheet

compatible versus incompatible elements very useful for testing whether rocks are related by fractional crystallisation processes.

g) Magma mixing occurs when different batches of magma are stirred together at depth is another important process that occurs in magma reservoirs beneath volcanoes. Calculate the Cr and Rb concentrations that you would obtain by mixing B1 and PT3 (the highest Rb sample) in the proportions 100:0, 75:25, 50:50, 25:75 and 0:100. Hint: set up two new columns with the different mixing proportions on a row and make use of the $ operator described in a) when selecting the element concentrations.

See Excel spreadsheet

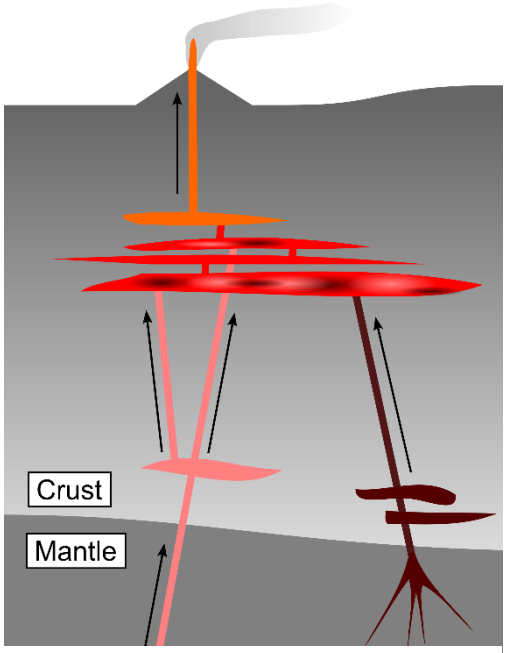


Figure 2: Cartoon showing where magma mixing can occur.

h) Show your results on the plot you produced for g) by right clicking on your plot and then left clicking on *Select Data.* From the *Select Data* dialogue click on *Add* to insert a second series by individually selecting data for x- and y-axis values in the boxes labelled *Series X values* and *Series Y values*, respectively. Note you will need to delete the ={1} shown in the *Series Y values* box first.

See Excel spreadsheet

i) How does the mixing trend differ from the fractionation trend?

The fractional crystallisation trend is strongly curved, whereas the magma mixing trend is linear between the basanite and phonotephrite endmembers. The data from La Palma show strong evidence for fractional crystallisation rather than mixing.

Question 2: Trace element fingerprints of different mantle sources (60 minutes)

- The trace element content of erupted magmas records information about the tectonic settings in which the magmas were produced; different mantle sources have different characteristic trace element 'fingerprints'. This is because elements show different partitioning behaviours during melting in different tectonic settings (Lecture 1 | Part 5).

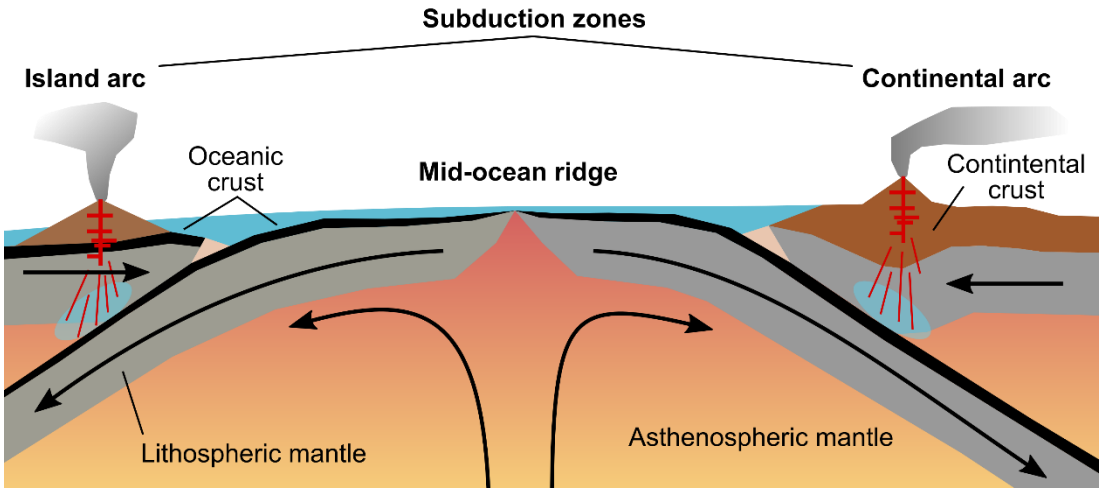


Figure 3: Comparing the tectonic settings in which mid-ocean-ridge and arc magmas are produced.

a) The sheet titled *Question 2* contains a data table with trace element compositions of important geochemical reservoirs (domains with distinct and internally coherent chemical characteristics) and two unknown basalts (basalt 1 and basalt 2). The elements are organised by their compatibility during dry mantle melting (i.e. at a mid-ocean ridge): Rb and Ba are highly incompatible while Yb and Y are relatively compatible.

b) Plot the compositions of both unknown basalts on a multi-element plot. Plot elements along the x-axis and concentrations on the y-axis. For the x-axis labels to work correctly, you will need to plot the data as a line graph instead of a scatter graph; select both the column of elements and a column of element concentrations from an unknown basalt (holding down *ctrl*) and select *Insert Line Chart → line* (do not plot as a stacked line plot).

c) Some elements are much more abundant than others, so change the y-axis from a linear axis to a logarithmic axis (by left then right clicking on the y-axis to open the *Format Axis* dialogue) and change where the x-axis crosses to an axis value of 0.01 so that the data do not overlap with the axis labels.

See Excel spreadsheet.

d) Add the composition of the depleted mantle to your plot. How does the composition of the depleted mantle compare with the compositions of the unknown basalts? Remember that more datasets can be added from the *Select Data* dialogue by right clicking on the plotted data like you did in Question 1.

See Excel spreadsheet. The concentration of most trace elements as about one order of magnitude lower in the depleted mantle than in the unknown basalts. However, highly incompatible trace elements are present in much higher in concentrations in one unknown basalt than in the other, meaning that concentrations of Rb and Ba, for example, differ between one and two orders of magnitude depending on the basalt with which the depleted mantle is being

e) Are the trace element compositions easy to interpret when plotted like this? Why do they look like they do (see Lecture 1 | Part 2)?

Trace element data are not easy to interpret when presented like this because of the odd-even abundance pattern described by the Oddo-Harkins rule that governs the solar system abundance of elements; elements with even numbers of protons are more stable and more abundant than those with odd numbers of protons.

f) In order to make trace element data easier to interpret, they are often plotted after being normalised to (i.e. divided by) the composition of a widely recognised geochemical reservoir like CI chondrite, bulk silicate Earth (BSE) or the depleted mantle. This results in much smoother patterns that are easier to interpret visually.

g) Normalise the two unknown basalt compositions to the depleted mantle composition (create some new columns of data for this) and show the normalised data in a new line plot. Adjust the axes as appropriate.

See Excel spreadsheet.

h) Compositional differences between the two unknown basalts should now be much clearer. Briefly describe these differences. For example, are the trace element patterns smooth? Do any elements show positive or negative anomalies? That is, are any elements present in much higher or lower concentrations than their neighbours?

Unknown basalt 1 has a smooth trace element pattern, with normalised trace element compositions being typically one order of magnitude greater than the depleted mantle. Highly incompatible element (Rb and Ba) are slightly depleted with respect to less incompatible elements.

Unknown basalt 2 has slightly irregular trace element pattern. Overall, highly incompatible elements are present at much greater abundances (Rb Ba, Th; two orders of magnitude greater than depleted mantle) than less incompatible elements (Yb, Y; less than one order of magnitude greater than depleted mantle). Ta and Nb, and possibly Hf and Zr, are present at lower levels than their neighbours (negative anomalies). Sr is present at higher levels (positive anomaly).

i) Some elements in one unknown basalt should be present at much lower concentrations than their neighbours. Which elements are they? Which geochemical grouping of elements do they belong to (Lecture 1 | Part 3)?

Ta and Nb, and possibly Hf and Zr, are present at lower levels than their neighbours in unknown basalt 2. They are high field strength elements (HFSE).

j) The two unknown basalts were erupted in different tectonic settings. Given what you know about the behaviour of the elements identified in i) during subduction, which basalt was erupted from an arc volcano? Where might the other basalt have erupted?

Unknown basalt 2 probably erupted in an arc because HFSE depletions are consistent with flux melting in the mantle source driven by the release of HFSE-poor aqueous fluids from subducting slabs. Unknown basalt 1 probably erupted at a mid-ocean ridge where the mantle source was dry. Moreover, it is relative depleted in highly incompatible trace elements, which suggests it did not erupt from an ocean island (i.e. hot spot) volcano.

k) Given your answer to j), can you explain the difference in the Rb and Ba contents of the two unknown basalts?

Unknown basalt 2 probably contains high concentrations of Rb and Ba, large ion lithophile elements, because they were transported in the aqueous fluids released from subducting slabs; unknown basalt 1 does not contain any slab-derived fluids.

l) Normalise the continental crust composition to the depleted mantle composition and plot the normalised data alongside the normalised data from the unknown basalts (add additional datasets via the *Select Data* dialogue). Based on the trace element pattern you observe, which tectonic environment appears to have played a key role in the creation of the continental crust?

See Excel spreadsheet. The continental crust shows a similar overall trace element pattern to arc basalts. Specifically, the continental crust is rich in LILE and poor in HFSE. It therefore seems highly likely that flux melting and arc magmatism in subduction zones have played a central role in the creation of the continental crust as we now see it.